Emily Howerton

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EDUCATION AND ACADEMIC APPOINTMENTS

Princeton University, Postdoctoral Research Associate (2024–Present)

Pennsylvania State University, PhD, Biology (2018–2023)
Dissertation: Quantitative Approaches to Improve the Management of Infectious Disease Outbreaks

College of Wooster, BA, Mathematics and Philosophy (2013–2017)
Summa Cum Laude; undergraduate thesis: On Rationality and Morality: Three Kinds of Approaches

PUBLICATIONS

Published and accepted (* denotes co-first authors)

1. SJ Bents, SL Larsen, S.-m Jung, **E Howerton**, et al. "Multi-model insights into racial and ethnic heterogeneities in early COVID-19 transmission and severity in the U.S." *American Journal of Epidemiology* (*Accepted*).
2. **E Howerton** and J Lessler. 2026. "Assessing model error in counterfactual worlds". *Journal of The Royal Society Interface* 23 242: p. 20251176. doi:10.1098/rsif.2025.1176.
3. J Sheen, **E Howerton**, HA Burrows, BL Rice, and CM Saad-Roy. 2026. "Challenges for the effectiveness of infectious disease interventions during global change". *Communications Health* 1 1: p. 34. doi:10.1038/s44528-026-00044-z.
4. W Benhamou, **E Howerton**, SW Park, C Viboud, CJE Metcalf, and BT Grenfell. 2026. "Leveraging perturbations to infer the population dynamics of human rhinovirus and interaction of influenza A virus". *PLOS Computational Biology* 22 9: e1014784. doi:10.1371/journal.pcbi.1014784.
5. L Hadley, S Ben Miled, HE Clapham, BA Djaafara, L Dyson, S Funk, JR Gog, H Heesterbeek, EM Hill, **E Howerton**, V Isham, A Kebir, et al. 2026. "How understanding the diversity of perspectives and systems in governments can increase the impact of scientific research". *Communications Health* 1 1: p. 13. doi:10.1038/s44528-026-00009-2.
6. M Kerr, R Borchering, AC Rivadeneira, L Contamin, S Funk, H Hochheiser, **E Howerton**, A Krystalli, L Shandross, and NG Reich. 2026. "A software platform for collaborative infectious disease modelling". *Nature Health*. doi:10.1038/s44360-026-00145-7.
7. L Shandross*, **E Howerton***, L Contamin, H Hochheiser, A Krystalli, CoIDM Hubs, NG Reich, and EL Ray. 2026. "Multi-Model Ensembles in Infectious Disease and Public Health: Methods, Interpretation, and Implementation in R". *Statistics in Medicine* 45 1-2: e70333. doi:10.1002/sim.70333.
8. RN Thompson, S Bansal, H Clapham, L Dyson, MA Gutierrez, L Hadley, WS Hart, H Heesterbeek, TD Hollingsworth, T House, **E Howerton**, V Isham, et al. 2026. "Infectious disease outbreak controllability: biological, social and public health factors". *Proceedings of the Royal Society B: Biological Sciences* 293 2063: p. 20252848. doi:10.1098/rspb.2025.2848.
9. SW Park, BF Nielsen, **E Howerton**, BT Grenfell, and S Cobey. 2026. "Susceptible host dynamics explain pathogen resilience to perturbations". *Proceedings of the National Academy of Sciences* 123 1: e2517518122. doi:10.1073/pnas.2517518122.
10. **E Howerton**, TC Williams, J.-S Casalegno, S Dominguez, R Gunson, K Messacar, CJE Metcalf, SW Park, C Viboud, and BT Grenfell. 2025. "Using COVID-19 pandemic perturbation to model RSV-hMPV interactions and potential implications under RSV interventions". *Nature Communications* 16 1: pp. 1–12. doi:10.1038/s41467-025-62358-w.
11. SW Park, I Holmdahl, **E Howerton**, W Yang, RE Baker, GA Vecchi, S Cobey, CJE Metcalf, and BT Grenfell. 2025. "Interplay between climate and childhood mixing can explain a sudden shift in RSV seasonality in Japan". *Nature Communications* 16 1: p. 11385. doi:10.1038/s41467-025-66184-y.
12. BF Nielsen, SW Park, **E Howerton**, OF Lorentzen, MH Jensen, and BT Grenfell. 2025. "Complex multiannual cycles of Mycoplasma pneumoniae: Persistence and the role of stochasticity". *Proceedings of the National Academy of Sciences* 122 52: e2509184122. doi:10.1073/pnas.2509184122.
13. S Loo, S.-m Jung, L Contamin, **E Howerton**, S Bents, et al. 2025. "Scenario projections of COVID-19 hospitalizations and deaths in response to immune escape levels and annual vaccination recommendations—United States, April 2024–April 2025". *JAMA Network Open* 8: e2532469. doi:10.1001/jamanetworkopen.2025.32469.

14. SW Park, B Noble, **E Howerton**, BF Nielsen, S Lentz, L Ambroggio, S Dominguez, K Messacar, and BT Grenfell. 2024"Predicting the impact of non-pharmaceutical interventions against COVID-19 on *Mycoplasma pneumoniae* in the United States". *Epidemics*: p. 100808. doi:10.1016/j.epidem.2024.100808.
15. **E Howerton**, T Langkilde, and K Shea. 2024. "Misapplied management makes matters worse: Spatially explicit control leverages biotic interactions to slow invasion". *Ecological Applications* n/a n/a: e2974. doi:10.1002/eap.2974.
16. LK Wade-Malone*, **E Howerton***, WJM Probert, MC Runge, C Viboud, and K Shea. 2024. "When do we need multiple infectious disease models? Agreement between projection rank and magnitude in a multi-model setting". *Epidemics* 47: p. 100767. doi:10.1016/j.epidem.2024.100767.
17. MC Runge, K Shea, **E Howerton**, K Yan, H Hochheiser, E Rosenstrom, WJM Probert, R Borchering, MV Marathe, B Lewis, S Venkatramanan, S Truelove, et al. 2024. "Scenario Design for Infectious Disease Projections: Integrating Concepts from Decision Analysis and Experimental Design". *Epidemics*: p. 100775. doi:10.1016/j.epidem.2024.100775.
18. S.-m Jung, SL Loo, **E Howerton**, L Contamin, CP Smith, EC Carcelén, K Yan, SJ Bents, J Levander, J Espino, JC Lemaitre, K Sato, et al. 2024. "Potential impact of annual vaccination with reformulated COVID-19 vaccines: Lessons from the US COVID-19 scenario modeling hub". *PLOS Medicine* 21 4: e1004387. doi:10.1371/journal.pmed.1004387.
19. C Bay, G St-Onge, JT Davis, M Chinazzi, **E Howerton**, J Lessler, MC Runge, K Shea, S Truelove, C Viboud, and A Vespignani. 2024. "Ensemble2: Scenarios ensembling for communication and performance analysis". *Epidemics* 46: p. 100748. doi:10.1016/j.epidem.2024.100748.
20. C Vanalli, **E Howerton**, F Yang, TN.-A Tran, and W Hu. 2024. "People and Data: solving planetary challenges together". *Frontiers in Environmental Science* 12. doi:10.3389/fenvs.2024.1332844.
21. SL Loo, **E Howerton**, L Contamin, CP Smith, RK Borchering, LC Mullany, S Bents, E Carcelen, S.-m Jung, T Bogich, WG van Panhuis, J Kerr, et al. 2024. "The US COVID-19 and Influenza Scenario Modeling Hubs: Delivering long-term projections to guide policy". *Epidemics* 46: p. 100738. doi:10.1016/j.epidem.2023.100738.
22. **E Howerton**, L Contamin, LC Mullany, M Qin, NG Reich, S Bents, RK Borchering, S.-m Jung, SL Loo, CP Smith, J Levander, J Kerr, et al. 2023. "Evaluation of the US COVID-19 Scenario Modeling Hub for informing pandemic response under uncertainty". *Nature Communications* 14 1: p. 7260. doi:10.1038/s41467-023-42680-x.
23. **E Howerton***, K Dahlin*, CJ Edholm, L Fox, M Reynolds, B Hollingsworth, G Lytle, M Walker, J Blackwood, and S Lenhart. 2023. "The effect of governance structures on optimal control of two-patch epidemic models". *Journal of Mathematical Biology* 87 5: p. 74. doi:10.1007/s00285-023-02001-8.
24. F Yang, TN.-A Tran, **E Howerton**, MF Boni, and JL Servadio. 2023. "Benefits of near-universal vaccination and treatment access to manage COVID-19 burden in the United States". *BMC Medicine* 21 1: p. 321. doi:10.1186/s12916-023-03025-z.
25. K Shea, RK Borchering, WJM Probert, **E Howerton**, TL Bogich, S.-L Li, WG van Panhuis, C Viboud, R Aguás, AA Belov, SH Bhargava, SM Cavany, et al. 2023. "Multiple models for outbreak decision support in the face of uncertainty". *Proceedings of the National Academy of Sciences* 120 18: e2207537120. doi:10.1073/pnas.2207537120.
26. FM Castonguay, JC Blackwood, **E Howerton**, K Shea, C Sims, and JN Sanchirico. 2023. "Optimal spatial evaluation of a pro rata vaccine distribution rule for COVID-19". *Scientific Reports* 13 1: p. 2194. doi:10.1038/s41598-023-28697-8.
27. **E Howerton**, MC Runge, TL Bogich, RK Borchering, H Inamine, J Lessler, LC Mullany, WJM Probert, CP Smith, S Truelove, C Viboud, and K Shea. 2023. "Context-dependent representation of within- and between-model uncertainty: aggregating probabilistic predictions in infectious disease epidemiology". *Journal of The Royal Society Interface* 20 198: p. 20220659. doi:10.1098/rsif.2022.0659.
28. RK Borchering, LC Mullany, **E Howerton**, M Chinazzi, CP Smith, M Qin, NG Reich, L Contamin, J Levander, J Kerr, J Espino, H Hochheiser, et al. 2023. "Impact of SARS-CoV-2 vaccination of children ages 5–11 years on COVID-19 disease burden and resilience to new variants in the United States, November 2021–March 2022: A multi-model study". *The Lancet Regional Health – Americas* 17: p. 100398. doi:10.1016/j.lana.2022.100398.
29. S Truelove, CP Smith, M Qin, LC Mullany, RK Borchering, J Lessler, K Shea, **E Howerton**, L Contamin, J Levander, J Salerno, H Hochheiser, et al. 2022. "Projected resurgence of COVID-19 in the United States in July-December 2021 resulting from the increased transmissibility of the Delta variant and faltering vaccination". *eLife* 11: e73584. doi:10.7554/eLife.73584.
30. **E Howerton**, MJ Ferrari, ON Bjørnstad, TL Bogich, RK Borchering, CP Jewell, JD Nichols, WJM Probert, MC Runge, MJ Tildesley, C Viboud, and K Shea. 2021. "Synergistic interventions to control COVID-19: Mass testing and isolation mitigates reliance on distancing". *PLOS Computational Biology* 17 10: e1009518. doi:10.1371/journal.pcbi.1009518.
31. JD Nichols, TL Bogich, **E Howerton**, ON Bjørnstad, RK Borchering, MJ Ferrari, M Haran, CP Jewell, K Pepin, WJM Probert,

JRC Pulliam, MC Runge, et al. 2021. "Strategic testing approaches for targeted disease monitoring can be used to inform pandemic decision-making". *PLOS Biology* 19 6: e3001307. doi:10.1371/journal.pbio.3001307.

Impact: USGS briefing to the head of the U.S. Department of the Interior and press release; reached more than 30,000 Facebook users and was reported by more than 10 news outlets.

32. RK Borchering, C Viboud, **E Howerton**, CP Smith, S Truelove, MC Runge, NG Reich, L Contamin, J Levander, J Salerno, W van Panhuis, M Kinsey, et al. 2021. "Modeling of Future COVID-19 Cases, Hospitalizations, and Deaths, by Vaccination Rates and Nonpharmaceutical Intervention Scenarios — United States, April–September 2021". *Morbidity and Mortality Weekly Report* 70 19: pp. 719–724. doi:10.15585/mmwr.mm7019e3.

Impact: Top 5% of research outputs scored by Altmetric; featured in a White House COVID-19 press briefing by CDC Director Rochelle Walensky; more than 550 tweets and coverage by more than 125 news outlets.

33. RD Pasteur, **E Howerton**, P Pozderac, S Young, and J Moore. 2018. "A flight-based metric for evaluating NFL punters". *Journal of Sports Analytics* 4 3: pp. 201–213. doi:10.3233/JSA-180164.

Submitted and in preparation

1. D DeAngelis, J Blackwood, O Gaoue, **E Howerton**, H Inamine, O Prosper, T Ramiadantsoa, K Shea, J Yeakel, M.-J Fortin, and NSF Theory Group. "New ecological theory to integrate internal ecological processes and external forces". *In revision*.
2. **E Howerton***, M Chung*, CJE Metcalf, BT Grenfell, and G Vecchi. "Could we predict infectious disease outbreaks like we predict the weather?" *Submitted*.
3. TS Johnston, R Subramanian, W Benhamou, **E Howerton**, R Schlesinger, et al. "Vaccine imprinting drives increased SARS-CoV-2 variant infection in children". *Submitted*.
4. W Oghenebrume, **E Howerton**, B Kampmann, BT Grenfell, and CJE Metcalf. "Measles and rubella reporting rates across 141 countries, 2010–2024: Comparison of aggregated and dynamic TSIR-based approaches". *Submitted*.
5. H Burrows, **E Howerton**, M Chung, M Kraemer, and CJE Metcalf. "Expect the unexpected: infectious disease in an era of global change". *Submitted*.
6. AA Nande, SL Larsen, J Turtle, JT Davis, SR Bandekar, B Lewis, S Chen, L Contamin, S.-m Jung, **E Howerton**, et al. "Assessing the impact of timing and coverage of United States COVID-19 vaccination campaigns: A multi-model approach". *Submitted*.
7. H Nguyen-Tran, **E Howerton**, LA Serebryannyy, AB Spaulding, SW Park, MR Vogt, P Permaul, E Ho, ML Hernandez, L Wang, M Castro, JA Bohl, et al. "Longitudinal immunoepidemiology of enterovirus D68 among young children in the US, 2022–2025: a pediatric multicenter, prospective, longitudinal cohort study". *Submitted*.
8. **E Howerton**, A Mahmud, DJD Earn, BT Grenfell, and CJE Metcalf. "Dynamic consequences of declining vaccination coverage in a heterogeneous world". *In preparation*.
9. S Fiandrino, D Paolotti, C Bay, M Chinazzi, JT Davis, **E Howerton**, et al. "Adaptive multi-model ensembles for improved epidemic projections and decision support". *In preparation*.

MATERIALS FOR PUBLIC & POLICY ENGAGEMENT

1. Coauthor on Scenario Modeling Hub reports provided to the U.S. Centers for Disease Control and Prevention (CDC) and other partners. Results supported decision-making, including U.S. CDC Advisory Committee on Immunization Practices recommendations for COVID-19 vaccination in children ages 5–11 and for multiple COVID-19 booster doses.
2. Invited to publish an article in *The Conversation* about research evaluating the impact of the U.S. Scenario Modeling Hub.

AWARDS

2018–2023	Inaugural Eberly College of Science Barbara McClintock Science Achievement Graduate Fellow
2023	Penn State University Alumni Association Dissertation Award
2022	Global Winner, Ernst & Young Building a Better Working World Data Challenge
2021	Peter J. Hudson Best Student Paper Award; Jeanette Ritter Mohnkern Graduate Student Scholarship in Biology
2020	National Science Foundation Graduate Research Fellowship, Honorable Mention
2018	University Graduate Fellowship; NCAA Postgraduate Scholarship Recipient

2017	National Science Foundation STEM Scholar (2015–2017); Women's Golf Coaches Association All-American Scholar (2015–2017); William A. Galpin Award for General Excellence in College Work; David A. Guldin Award in Physical Education; John F. Miller Prize in Philosophy; William H. Wilson Prize in Mathematics
2016	Edward Taylor Prize in Mathematics
2015	Andrew D. Cronin Emerging Leader Award; Lyman C. Knight Sr. Prize in Mathematics and Physical Education

SERVICE AND LEADERSHIP

Princeton University

2025–Present	Member, *Epidemics* Editorial Advisory Board
2024–2025	Member, *Epidemics*[10] Scientific Advisory Committee
2024–Present	Member, Ecology and Evolutionary Biology Climate Committee for All
2024, 2025	Mentor and Application Review Committee Member, EEB Scholars Program

Pennsylvania State University

2020–2023	Member, Eberly College of Science Climate and Diversity Committee
2022	Project mentor, MIDAS Network/Harvard CCDD Conference to Increase Diversity in Mathematical Modeling and Public Health, including a pre-conference workshop on effective teaching practices
2020–2021	Co-president, Center for Infectious Disease Dynamics Graduate Student Association
2019	Co-designer and co-leader, Girl Scout Workshop; mentor, First Generation Advocates, Penn State University; volunteer, Discovery Space, State College, Pennsylvania

GRANTS AND FUNDING

Center for Infectious Disease Dynamics Graduate Student Association Travel Grant. 2021. $1,000.

National Science Foundation Research Experience for Undergraduates Supplement. May 1–August 1, 2022. $7,000. Supplement to the EEID grant "US–UK Collab: Adaptive surveillance and control for endemic disease elimination" (PI: Matthew J. Ferrari). With Katriona Shea (co-PI), advised La Keisha Wade-Malone in writing the proposal to fund a second summer of her research project.

National Science Foundation Research Experience for Undergraduates Supplement. May 1–August 1, 2020. $6,000. Supplement to the EEID grant "US–UK Collab: Adaptive surveillance and control for endemic disease elimination" (PI: Matthew J. Ferrari). With Matthew J. Ferrari and Katriona Shea (co-PI), helped write the grant and served as the named graduate student mentoring an undergraduate researcher.

Huck Institutes of the Life Sciences Coronavirus Research Seed Fund. March 11, 2020–March 10, 2021. $73,785. "Embracing uncertainty in COVID-19 management" (PI: Katriona Shea). Co-wrote the proposal with Katriona Shea and served as the named graduate student.

PRESENTATIONS (* DENOTES PRESENTER)

Invited Oral Presentations

1. Models of infectious disease dynamics: Making predictions to support public health response. **E Howerton***. *Department of Biology and Environmental Data Science Seminar, James Madison University*. April 11, 2025.
2. Multi-model scenario projections to support public health in real time: Challenges and opportunities from the U.S. Scenario Modeling Hubs. **E Howerton***. *Disease Decision Analysis & Research Group, U.S. Geological Survey*. February 10, 2025.
3. Informing public health response in the face of uncertainty: Lessons from the U.S. Scenario Modeling Hub. **E Howerton***, C Viboud, and J Lessler on behalf of the COVID-19 Scenario Modeling Hub. *Isaac Newton Institute, Modelling and Inference for Pandemic Preparedness—A Focused Workshop*. August 5–9, 2024. Recording.
4. Synergistic interventions to control COVID-19. **E Howerton***, MJ Ferrari, ON Bjørnstad, TL Bogich, RK Borchering, CP Jewell, JD Nichols, WJM Probert, MJ Tildesley, MC Runge, C Viboud, and K Shea. *Joint Mathematics Meetings*. April 6–8, 2022.
5. Synergistic interventions to control COVID-19. **E Howerton***, MJ Ferrari, ON Bjørnstad, TL Bogich, RK Borchering, CP Jewell, JD Nichols, WJM Probert, MJ Tildesley, MC Runge, C Viboud, and K Shea. *Department of Biology, Pennsylvania State University*. Invited presentation by the Jeanette Ritter Mohnkern Award winner. August 24, 2021.

Oral Presentations

1. The honeymoon is over: Dynamic impacts of abrupt reductions in vaccine uptake. **E Howerton***, BT Grenfell, CJE Metcalf. *Epidemics*[10]. December 1, 2025.
2. Community dynamics of human respiratory pathogens: The HMPV story. **E Howerton***. *Department of Ecology and Evolutionary Biology, Princeton University*. February 7, 2025.
3. Modeling the impact of RSV interventions on hMPV burden and dynamics. **E Howerton***. *Fogarty International Center EPS Modeling Seminar, National Institutes of Health*. October 16, 2024.
4. An introduction to human metapneumovirus. **E Howerton***. *Life Course Immunity: A Multi-Pathogen Perspective*. February 9, 2024.
5. Informing pandemic response in the face of uncertainty: An evaluation of the U.S. COVID-19 Scenario Modeling Hub. **E Howerton***, C Viboud, and J Lessler on behalf of the COVID-19 Scenario Modeling Hub. *Epidemics*[9]. November 28, 2023.
6. What does it mean to evaluate a scenario projection? Lessons from the COVID-19 Scenario Modeling Hub. **E Howerton***, J Lessler, and C Viboud on behalf of the COVID-19 Scenario Modeling Hub contributors. *Royal Society Meeting on Forecasting Infectious Disease Incidence*. March 15, 2023.
7. Collaborative open hubs for infectious disease modeling. C Viboud*, **E Howerton**, A Vespignani, and J Lessler on behalf of the COVID-19 Scenario Modeling Hub and Ebola Forecast Challenge contributors. *Royal Society Meeting on Forecasting Natural and Social Systems*. March 14, 2023.
8. Optimizing management decisions for control of infectious disease outbreaks. **E Howerton***. *Northeastern University*. February 6, 2023.
9. A look back at fifteen rounds of SMH: Evaluation. **E Howerton***. *COVID-19 Scenario Modeling Hub Meeting 2022—Past, Present, and Future of Scenario Modeling for the U.S.* September 14, 2022.
10. Looking back at one year of the COVID-19 Scenario Modeling Hub. **E Howerton***, on behalf of the COVID-19 Scenario Modeling Hub Team. *MIDAS Annual Meeting*. September 8, 2022.
11. Misapplied management makes matters worse: Spatially explicit control strategies that leverage biotic interactions may more effectively prevent invader spread. **E Howerton***, T Langkilde, and K Shea. August 17, 2022.
12. How to get the best out of multiple disease models. **E Howerton***, MC Runge, TL Bogich, RK Borchering, H Inamine, J Lessler, LC Mullany, WJM Probert, CP Smith, S Truelove, C Viboud, and K Shea. *Ecology and Evolution of Infectious Diseases*. June 6–9, 2022. Recording.
13. Scenario Modeling Hub round 13 preview and evaluation. J Lessler* and **E Howerton*** on behalf of the COVID-19 Scenario Modeling Hub. *U.S. Centers for Disease Control and Prevention Council of State and Territorial Epidemiologists*. April 5, 2022.
14. Comparing aggregation methods for the COVID-19 Scenario Modeling Hub. **E Howerton***. *COVID-19 Scenario Modeling Hub Weekly Meeting*. June 18, 2021.

Poster Presentations

1. Serology illuminates drivers of early-life respiratory virus transmission dynamics: The PREMISE study (2022–2025). **E Howerton***, H Nguyen-Tran[†], L Serebryannyy[†], AB Spaulding, SW Park, MR Vogt, P Permaul, EC Ho, ML Hernandez, M Miller, L Wang, M Castro, JA Bohl, S Godbole, JA Liao, CJE Metcalf, SR Dominguez, DW Kimberlin, K Messacar[‡], DC Douek[‡], BT Grenfell. *Northeastern Ecology and Evolution of Infectious Diseases*, Cornell University. November 6–7, 2026. [†]Co-first authors; [‡]co-senior authors.
2. Dynamic consequences of declining vaccination coverage in a heterogeneous world. **E Howerton***, A Mahmud, DJD Earn, BT Grenfell, CJE Metcalf. *Ecology and Evolution of Infectious Disease Dynamics*. June 3, 2026.
3. Cross-protection and the lags between RSV and human metapneumovirus outbreaks. **E Howerton***, TC Williams, J-S Casalegno, CJE Metcalf, SW Park, BT Grenfell. *Ecology and Evolution of Infectious Disease Dynamics*. June 25, 2024.
4. Informing pandemic response in the face of uncertainty: An evaluation of the U.S. COVID-19 Scenario Modeling Hub. **E Howerton***, K Shea, C Viboud, and J Lessler on behalf of the COVID-19 Scenario Modeling Hub. *Ecology and Evolution of Infectious Disease Dynamics*. May 23, 2023.
5. Theory of combining probabilistic projections with applications in epidemiology. **E Howerton***, MC Runge, TL Bogich, RK Borchering, H Inamine, J Lessler, LC Mullany, WJM Probert, CP Smith, S Truelove, C Viboud, and K Shea. *Epidemics*[8]. November 29–December 1, 2021.

6. Synergistic interventions to control COVID-19: Mass testing and isolation mitigates reliance on distancing. **E Howerton***, MJ Ferrari, ON Bjørnstad, TL Bogich, RK Borchering, CP Jewell, JD Nichols, WJM Probert, MJ Tildesley, MC Runge, C Viboud, and K Shea. *Ecology and Evolution of Infectious Diseases*. June 14–17, 2021.

TEACHING

Pennsylvania State University

TA	Ecological and Environmental Problem Solving	Spring 2020, Spring 2021

College of Wooster

TA	First Year Seminar	Fall 2014, Fall 2015
TA	Calculus with Algebra A	Fall 2014, Fall 2015, Fall 2016
TA	Calculus with Algebra B	Spring 2015, Spring 2016, Spring 2017
TA	Logic & Philosophy	Spring 2015
Tutor	Math Center (Precalculus–Calculus III)	Spring 2015–Spring 2017

WORKSHOP PARTICIPATION

Aug. 2026	University of Vermont, "Workshop on Public Health Policy, Risk Perception, and Infectious Disease," August 10–30
2024, 2025	Isaac Newton Institute, "Modelling and Inference for Pandemic Preparedness—A Focused Workshop," University of Cambridge, August 5–9, 2024, and December 8–19, 2025
May 2024	Communication Arts & Sciences Summer Symposium, "Viral Movements," Pennsylvania State University, May 15–16
Mar. 2023	The Royal Society Satellite Meeting, "Forecasting Infectious Disease Incidence for Public Health," London, United Kingdom, March 15–16
June 2022	Ecology and Evolution of Infectious Disease Dynamics Workshop, "Pandemic Scenario Modeling and Science Communication," June 3–6
May–June 2021	American Mathematical Society Mathematics Research Community, "Dynamics of Infectious Diseases: Ecological Models Across Multiple Scales," May 30–June 5
Oct. 2019	National Science Foundation, "Advancing Ecological Theory" Workshop, Pennsylvania State University, October 23–25

PROFESSIONAL MEMBERSHIPS

Ecological Society of America; American Association for the Advancement of Science; American Mathematical Society; Phi Beta Kappa Academic Honor Society.

REVIEWING ACTIVITY

American Journal of Epidemiology; BMC Global Health; BMC Medical Informatics and Decision Making; BMC Public Health; Communications Biology; eLife; Epidemics; Epidemiology and Infection; Journal of Theoretical Biology; Lancet Digital Health; Lancet Regional Health—Americas; Mathematical Biosciences; Mathematical Biosciences and Engineering; PLOS Computational Biology; PLOS ONE; Journal of the Royal Society Interface; Royal Society Open Science; Science Advances; Scientific Reports.